COMPUTER VISION

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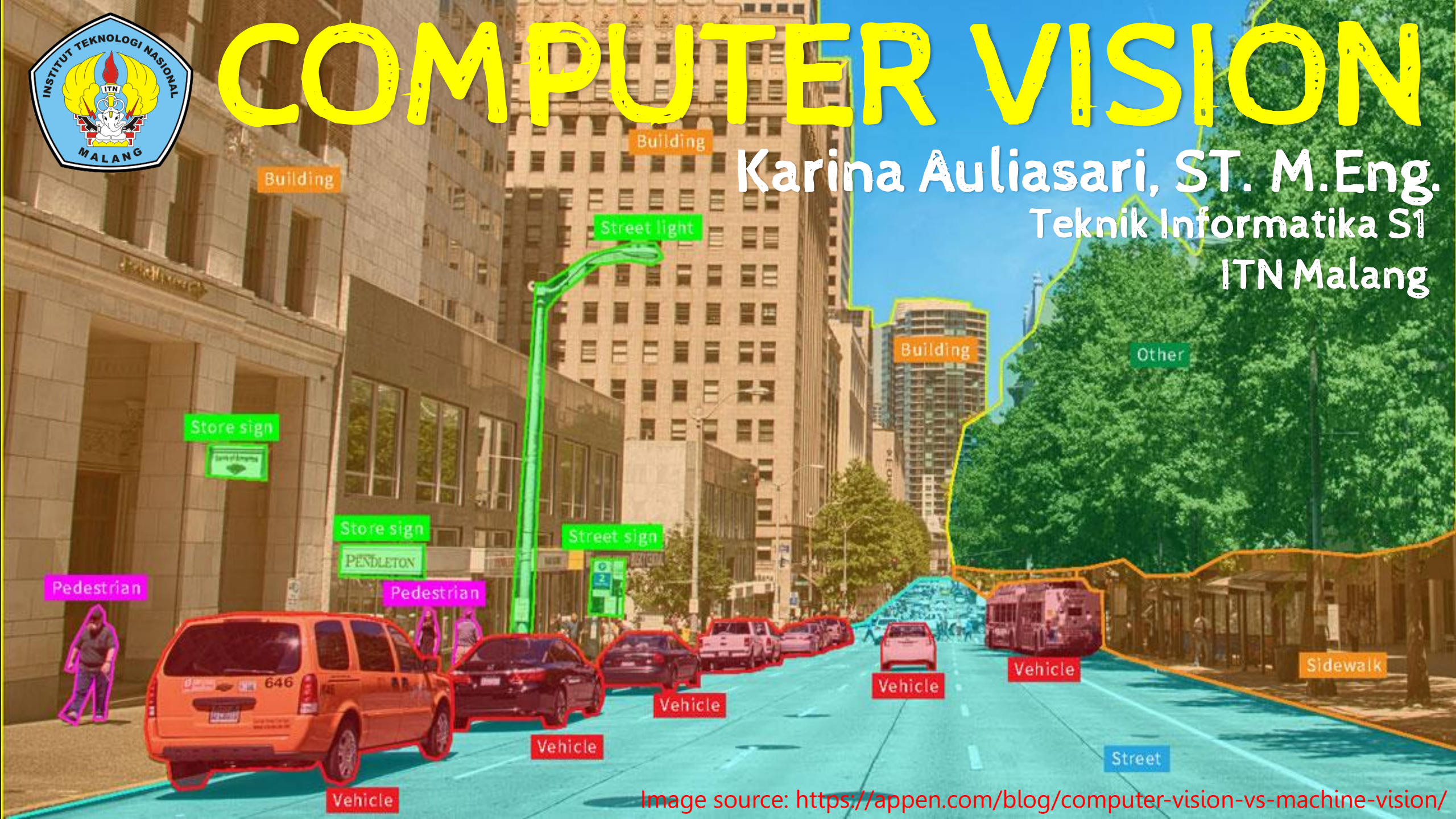

Image source: https://appen.com/blog/computer-vision-vs-machine-vision/

COURSE OUTLINE:

1. Image Pre-processing
2. Image Pre-processing with Pillow Library

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01 Image Pre-processing

- **Tujuan** dari proses *Image Preprocessing* adalah untuk **meningkatkan kualitas fitur citra** yang nantinya fitur tersebut kita proses lebih lanjut di proses *feature extraction.*
- Beda studi kasus maska berbeda proses *image preprocessing* nya.
- Contoh: Jika kita mengembangkan aplikasi deteksi manusia dan menghitung objek manusia di suatu citra, maka kita akan fokus pada objek manusia pada citra tersebut untuk diotomatisasi lebih lanjut oleh karena itu background pemandangan, jalan, tembok atau objek lain tidak menjadi fokus kita, oleh karena itu perlu ada proses *Image Preprocessing* untuk menonjolkan objek manusia di suatu citra.

02 Beberapa Contoh *Image Pre-processing*

- Rotasi,
- Flipping,
- Shifting,
- Scaling,
- Cropping,
- Filtering,
- Perubahan ruang warnacitra,
- Etc.

03 Rotation

```
[2] from google.colab import drive
    drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=T
```

```
[3] cd 'drive/MyDrive/Colab Notebooks'
```

```
/content/drive/MyDrive/Colab Notebooks
```

```
[4] #Import Library for Image Preprocessing
    from PIL import Image, ImageOps, ImageChops, ImageFilter
    import matplotlib.pyplot as plt
```

```
image = Image.open('walking_people.jpg')
plt.imshow(image)
```

```
#Rotation
#background
white=(255,255,255)
#arah rotasi -30
spinPicture030 = image.rotate(-30, fillcolor=white)

#spinPicture090 = image.rotate(90)
spinPicture030.save('walking_people_30.jpg')
plt.imshow(spinPicture030)
```

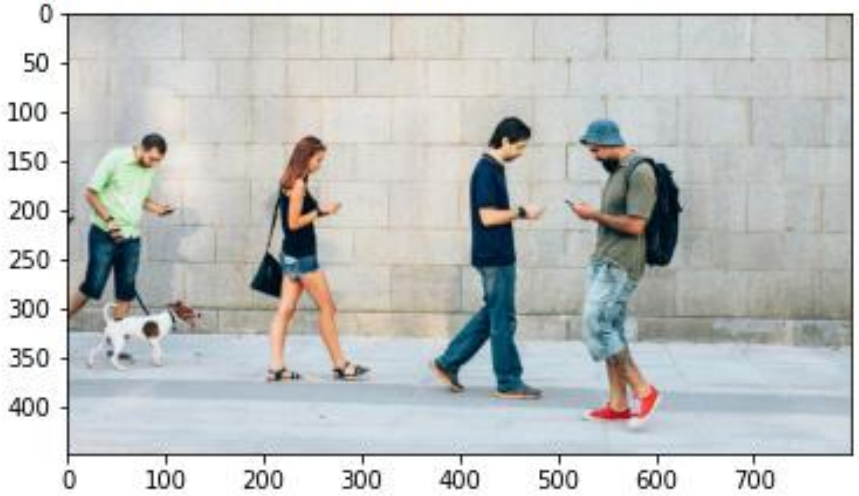
04 Flipping

```
#Flipping
##Vertical Flip
im_flip = ImageOps.flip(image)
im_flip.save('walking_people_vertical_flip.jpg', quality=95)
plt.imshow(im_flip)
```

```
<matplotlib.image.AxesImage at 0x7fe43503ee50>
```

```
##Horizontal Flip
im_flip = ImageOps.mirror(image)
im_flip.save('walking_people_horizontal_flip.jpg', quality=95)
plt.imshow(im_flip)
```

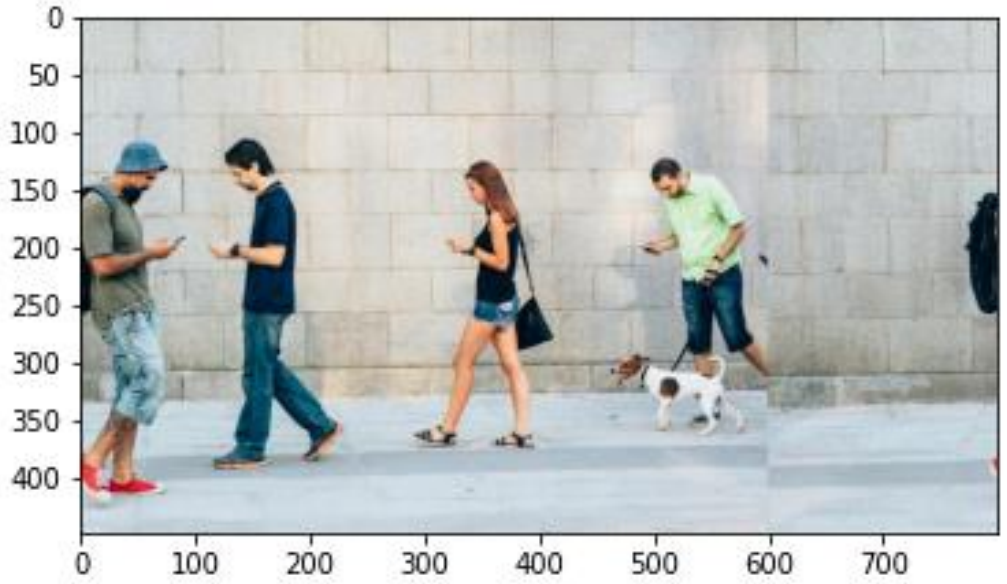
```
<matplotlib.image.AxesImage at 0x7f3cdebf2e50>
```



05 Shifting

```
#Shifting (Pergeseran Piksel)
##shifting -200 di sumbu x
im_shifting_200_left = ImageChops.offset(image, -200, 0)
im_shifting_200_left.save('walking_people_shifting_200_left.jpg')
plt.imshow(im_shifting_200_left)
```

```
<matplotlib.image.AxesImage at 0x7f3cdea67f10>
```



05 Shifting

```
##shifting -200 di sumbu y left 200 up
im_shifting_200_left = ImageChops.offset(image, -200, -200)
im_shifting_200_left.save('walking_people_shifting_200_left_200_up.jpg')
plt.imshow(im_shifting_200_left)
```

```
<matplotlib.image.AxesImage at 0x7f3cde9e3ad0>
```

06 Scaling

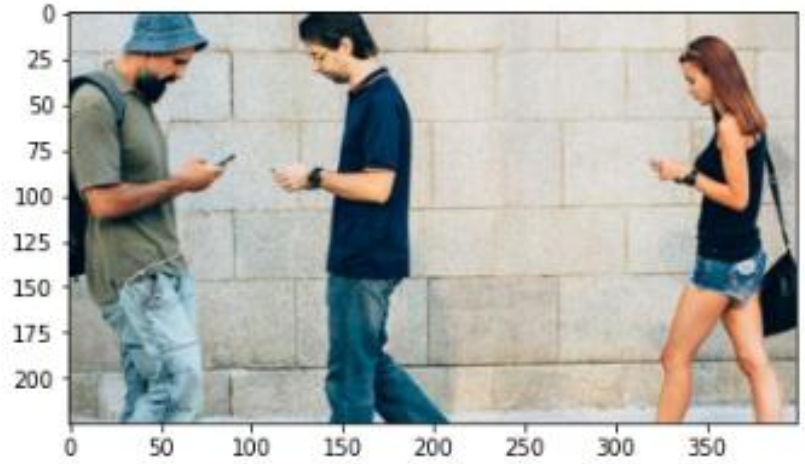
```
#Scaling
image_scaling = image.resize((1024,512))
image_scaling.save('walking_people_resize_image.jpg')
plt.imshow(image_scaling)
```

```
<matplotlib.image.AxesImage at 0x7f3cdea76690>
```

06 Cropping

```python
#Cropping
#Get Dimensions
width, height = image.size
print(width, " ",height)
#x1,y1
left = width/4
top = height/4
#x2,y2
right = 3 * width/4
bottom = 3 * height/4
image_cropped = image.crop((left, top, right, bottom))
print(image_cropped.size)
image_cropped.save('image_cropped.jpg')
plt.imshow(image_cropped)
```

```
800   449
(400, 225)
<matplotlib.image.AxesImage at 0x7fe42b987b50>
```



TERIMA KASIH

SEMOGA BERMANFAAT